

LI Wanyue

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Education

The University of Hong Kong Ph.D. in Mechanical Engineering, Department of Engineering	2024/02 – Present Hong Kong
Sun Yat-sen University M.S. in Artificial Intelligence, School of Artificial Intelligence	2020/09 – 2023/06 Guangzhou
South China Agricultural University B.S. in Computer Science and Technology, College of Mathematics and Informatics	2015/09 – 2019/06 Guangzhou

Publications

Like Playing a Video Game: Spatial-Temporal Optimization of Foot Trajectories for Controlled Football Kicking in Bipedal Robots

Wanyue Li, Ji Ma, Minghao Lu and Peng Lu*
2025 *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025 Best Paper Award)*.

Dynamic Locomotion of a Quadruped Robot with Active Spine via Model Predictive Control

Wanyue Li, Zida Zhou, Hui Cheng*
2023 *IEEE International Conference on Robotics and Automation (ICRA 2023)*.

MGDP: Mastering a Generalized Depth Perception Model for Quadruped Locomotion

Yinzhaodong, Ji Ma, Yidan Lu, Jiahui Zhang, Wanyue Li, Yeke Chen, Xuechao Chen, Zhangguo Yu, and Peng Lu
2026 *Advanced Science (Adv. Sci. 2026)*.

MILD: Tractable Terrain Modeling for Learning Improved Bipedal Locomotion on Deformable Surfaces

Zeren Luo, Jiahui ZHANG, Zhe Xu, Wanyue Li, Xinqi Li, Xuechao Chen, Zhangguo YU, Annan Tang, Peng Lu*.
2025 *IEEE Robotics and Automation Letters (RA-L 2025)*.

MARG: MAstering Risky Gap Terrains for Legged Robots with Elevation Mapping

Yinzhaodong, Ji Ma, Liu Zhao, Wanyue Li, Peng Lu*
2025 *IEEE Transactions on Robotics (T-RO 2025)*.

RM-Planner: Integrating Reinforcement Learning with Whole-Body Model Predictive Control for Mobile Manipulation

Zixuan Zhuang, Le Zheng, Wanyue Li, Renming Liu, Peng Lu, Hui Cheng*.
2024 *IEEE International Conference on Robotics and Automation (ICRA 2024)*.

Experience

XPENG Robotics Motion Control Algorithm Developer Fully participated in the development of motion control algorithms for XPENG's first-generation bipedal robot, PX5.	2023/07 – 2024/02
PNDbotics Robotics Algorithm Research Intern Developing dynamics-based interactive motion retargeting algorithms for humanoid robots.	2026/04 – Present

Honors and Awards

IROS 2025 (Best Paper Award)	2025/10
ATEC 2025 Tech Elite Competition (5th Place)	2025/08
1st "Yat-sen Warrior Cup" Collaborative Robotics International Invitational (Second Prize)	2023/08

Skills

- **English:** IELTS 6.5
- **Programming:** C/C++, Python, MATLAB
- **Frameworks:** ROS/ROS2, Isaac Lab / Isaac Gym, MuJoCo, Pinocchio, CasADi
- **Algorithms:** Imitation Learning, Reinforcement Learning, Retargeting, Model Predictive Control, Trajectory Optimization